

Explainable Causal ST-GNN Digital Twin for Real-Time Air Pollution Forecasting and Source Attribution.

● 1. Core Problem

Current AQI systems only display pollution levels.

They cannot:

Predict future AQI reliably.

Identify which source caused pollution spikes.

Simulate “what-if” policies.

Quantify prediction confidence.

Authorities lack intelligent tools for intervention planning.

● 2. Our Main Goal

We build a City Digital Twin for Air Pollution that:

- ✓ Forecasts AQI (1h, 3h, 6h ahead)
- ✓ Identifies pollution source contributions
- ✓ Learns causal relationships between variables
- ✓ Simulates intervention scenarios
- ✓ Recommends optimal mitigation strategies
- ✓ Provides uncertainty-aware predictions

● 3. What is a Digital Twin?

A digital twin is a virtual replica of a real-world system.

Here, the city becomes a dynamic graph model.

It continuously learns from real-time data.

It can simulate future pollution outcomes.

● 4. How We Model the City

Nodes:

Air Quality Monitoring Stations

Edges:

Geographic proximity

Wind-direction flow

Road connectivity

Node Features:

PM2.5, PM10, NO₂

Temperature

Humidity

Wind speed

Wind direction

Traffic intensity proxy

● 5. System Architecture

Module 1: Spatio-Temporal Forecasting

Graph Attention Network (spatial dependency)

Temporal Attention (time-series modeling)

Multi-horizon AQI forecasting

Output: Future AQI values

Module 2: Source Attribution

Separate feature groups:

Traffic-related

Meteorological transport

Attention weights used to compute contribution percentage.

Output: Traffic $\rightarrow$ X%

Wind transport $\rightarrow$ Y%

Local weather $\rightarrow$ Z%

Module 3: Causal Discovery

Use causal inference algorithms (e.g., NOTEARS/PCMCI).

Learn cause-effect relationships between:

Traffic

Wind

AQI

Industrial indicators

Output: Causal graph explaining pollution dynamics.

Module 4: Counterfactual Simulation

We simulate:

30% traffic reduction

Wind pattern shift

Industrial emission control

Predict:

“How would AQI change if intervention applied?”

Module 5: Reinforcement Learning Policy Layer

RL agent learns optimal pollution mitigation strategy.

Reward = AQI reduction with minimal disruption.

Output: Recommended intervention policy.

Module 6: Uncertainty Estimation

Monte Carlo Dropout.

Provides confidence interval.

Example: AQI = 170 ± 12

● 6. Evaluation

Forecast Metrics:

RMSE

MAE

R^2

Causal Metrics:

Counterfactual consistency

Causal discovery accuracy

Policy Simulation:

AQI reduction percentage

Stability under weather variations

● 7. Why This Is Strong

Combines:

Graph Neural Networks

Causal AI

Explainable AI
Digital Twin modeling
Reinforcement Learning
Uses real-world Indian data.
Supports smart city governance.
Research-level contribution.

● 8. Social Impact

Helps pollution control boards.
Enables data-driven traffic policies.
Supports sustainable urban planning.
Improves public health decision-making.